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DEPARTMENT OF CSE

ASSIGNMENT

COURSE NAME : Electrical Circuits

COURSE CODE : 0713-101

ASSIGNMENT NO. : 01

ASSIGNMENT NAME : Comprehensive Analysis of capacitor Behavior
in DC and AC Circuits : A Problem Solving
Approach.

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Title : Comprehensive Analysis of Capacitor Behavior in DC and AC Circuits : A Problem Solving Approach .

Objectives:

1. To analyze and solve problems related to capacitor behavior in both DC and AC circuits.
2. To demonstrate the calculation of equivalent resistance in series and parallel configurations
3. To determine energy stored (w) in capacitive circuits .
4. To analyze current, voltage drop, charge and time relationships in capacitor operations.
5. To develop systematic problem-solving approaches for capacitive circuit analysis.

Introduction :

In electrical engineering, capacitors serve as fundamental components that store and release electrical energy, playing a crucial role in both DC and AC circuits. This assignment presents a systematic analysis of '20' capacitor related problems, encompassing various aspects of capacitor behavior and characteristics. The problems explore key concepts including energy storage, equivalent capacitance in series and parallel configurations, current, voltage drop, charge, time relationship. Through detailed problem solving, this work demonstrates the practical application of capacitor principle in both direct current and alternating current scenarios, providing a thorough understanding of capacitive circuit analysis.

Problem : 01

- a) calculate the charge stored on a 3pF with 20V across it.
b) Find the energy stored in the capacitor.

$\Rightarrow$ Solution :

a) Given, $C = 3 \times 10^{-12} \text{ F}$

$$V = 20 \text{ V}$$

$$\therefore q = CV = 3 \times 10^{-12} \times 20 = 60 \times 10^{-12} \text{ C} \quad (\text{Ans})$$

$$\begin{aligned} \text{b) } W &= \frac{1}{2} CV^2 = \frac{1}{2} \times 3 \times 10^{-12} \times 20^2 \\ &= 6 \times 10^{-10} \text{ J} \end{aligned}$$

Problem : 02 - What is the voltage Across a $4.5 \mu\text{F}$ Capacitor if the charge on one plate is 0.12 mC ? How much energy is stored?

$\Rightarrow$ Solution : Given,

$$C = 4.5 \times 10^{-6} \text{ F}$$

$$q = 0.12 \times 10^{-3} \text{ C}$$

$$q = CV$$

$$\therefore V = \frac{q}{C} = \frac{0.12 \times 10^{-3}}{4.5 \times 10^{-6}} = 26.67 \text{ V}$$

$$\begin{aligned} \therefore W &= \frac{1}{2} CV^2 = \frac{1}{2} \times 4.5 \times 10^{-6} \times (26.67)^2 \\ &= 1.6 \times 10^{-3} \text{ J} \end{aligned}$$

(Ans)

Problem : 3 - The voltage across $5\text{-}\mu\text{F}$ capacitor is $v(t) = 10 \cos 6000t$ V. Calculate the current through it.

Solution $\Rightarrow$

Given,

$$C = 5 \times 10^{-6} \text{ F}$$

We know,

$$i = C \frac{dv}{dt}$$

$$= 5 \times 10^{-6} \frac{d}{dt} (10 \cos 6000t)$$

$$= 5 \times 10^{-6} \times 10 \times (-\sin 6000t) \times 6000$$

$$= -0.3 \sin 6000t \text{ A} \quad (\text{Ans})$$

Problem : 4 - If a $10\text{-}\mu\text{F}$ capacitor is connected to a voltage source with $v(t) = 75 \sin 2000t$ V. Determine the current through the capacitor.

Solution $\Rightarrow$ Given, $C = 10 \times 10^{-6} \text{ F}$

$$\text{we know, } i = C \frac{dv}{dt}$$

$$= C \frac{d}{dt} (75 \sin 2000t)$$

$$= 10 \times 10^{-6} \times 75 \times (\cos 2000t) \times 2000$$

$$= 1.5 \cos 2000t$$

(Ans.)

Problem : 5 - Determine the voltage across a $2\mu\text{F}$ capacitor if the current through it is $i(t) = 6e^{-3000t} \text{ mA}$. Assume the initial capacitor voltage is zero.

Solution :

Given,

$$C = 2 \times 10^{-6} \text{ F}$$

$$i(t) = 6e^{-3000t} \text{ mA}$$

$$V(0) = 0$$

We know,

$$V = \frac{1}{C} \int_0^t i dt$$

$$= \frac{1}{2 \times 10^{-6}} \int_0^t 6 \times 10^{-3} e^{-3000t} dt$$

$$= \frac{6 \times 10^{-3}}{2 \times 10^{-6}} \int_0^t e^{-3000t} dt$$

$$= \frac{6 \times 10^{-3}}{2 \times 10^{-6}} \left[-\frac{e^{-3000t}}{3000} \right]_0^t$$

$$= \frac{6 \times 10^{-3}}{2 \times 10^{-6} \times 3000} \left[-e^{-3000t} + e^{-3000 \times 0} \right]$$

$$= (1 - e^{-3000t}) \text{ V}$$

(Ans.)

Problem 6: The current through a $100 \mu\text{F}$ capacitor is $i(t)$,
 $i(t) = 50 \sin 120\pi t \text{ mA}$. Calculate the voltage across it at
 $t = 1 \text{ ms}$ and $t = 5 \text{ ms}$. Take $v(0) = 0$.

Solution:

Given,

$$C = 100 \times 10^{-6} \text{ F}$$

$$i(t) = 50 \sin 120\pi t \text{ mA}$$

$$v(0) = 0$$

$$\begin{aligned}\therefore v(t) &= \frac{1}{C} \int_0^t i \, dt + v_0 \\&= \frac{1}{100 \times 10^{-6}} \int_0^t 50 \times 10^{-3} \sin 120\pi t \, dt + 0 \\&= \frac{50 \times 10^{-3}}{100 \times 10^{-6}} \int_0^t \sin 120\pi t \, dt - \\&= \frac{50 \times 10^{-3}}{100 \times 10^{-6}} \left[\frac{-\cos 120\pi t}{120\pi} \right]_0^t \\&= \frac{50 \times 10^{-3}}{100 \times 10^{-6} \times 120\pi} (-\cos 120\pi t + \cos 120\pi \times 0) \\&= \frac{25}{6\pi} (-\cos 120\pi t + 1)\end{aligned}$$

$$\therefore t = 1 \times 10^{-3} \text{ s}$$

$$\therefore v = \frac{25}{6\pi} (-\cos 120\pi \times 1 \times 10^{-3} + 1) = 0.09314 \text{ V}$$

$$\therefore t = 5 \times 10^{-3} \text{ s}$$

$$\therefore v = \frac{25}{6\pi} (-\cos 120\pi \times 5 \times 10^{-3} + 1) = 1.736 \text{ V}$$

(Ans)

Problem 7: What is the voltage across a $3 \mu\text{F}$ capacitor if the charge on one plate is 0.12 mC ? How much energy stored?

Solution $\Rightarrow$

Given,

$$C = 3 \times 10^{-6} \text{ F}$$

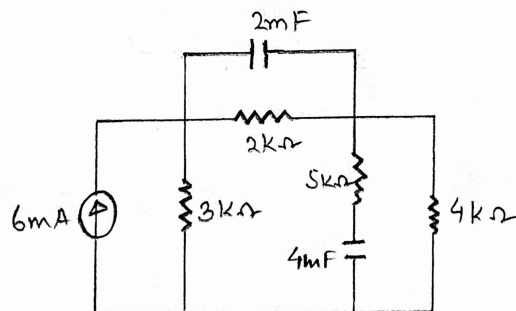
$$q = 0.12 \times 10^{-3} \text{ C}$$

$$\therefore V = \frac{q}{C} = \frac{0.12 \times 10^{-3}}{3 \times 10^{-6}} = 40 \text{ V}$$

$$\therefore W = \frac{1}{2} CV^2 = \frac{1}{2} \times 3 \times 10^{-6} \times (40)^2 = 2.4 \times 10^{-3} \text{ J}$$

(Ans)

Problem 8: Obtain the energy stored in each capacitor under DC condition.



Fig

Solution $\Rightarrow$

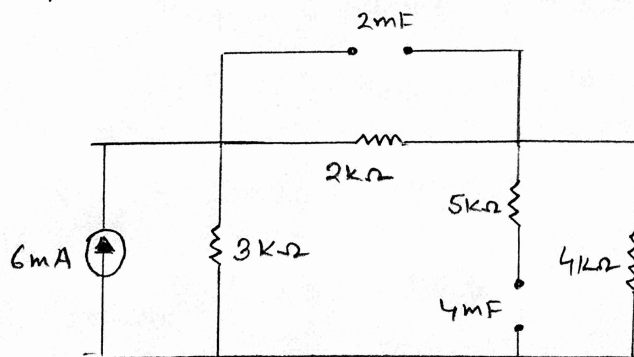


Fig: 1

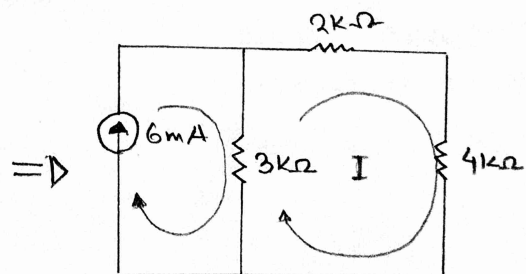


Fig: 2

Applying KVL at mesh I in fig: 2,

$$2I + 4I + 3(I - 6) = 0$$

$$\Rightarrow 8I - 18 = 0 \quad \therefore I = 2$$

Here, All capacitors replaced by open circuit under dc condition.

$$\therefore V_{2k\Omega} = 2 \times 2 = 4V$$

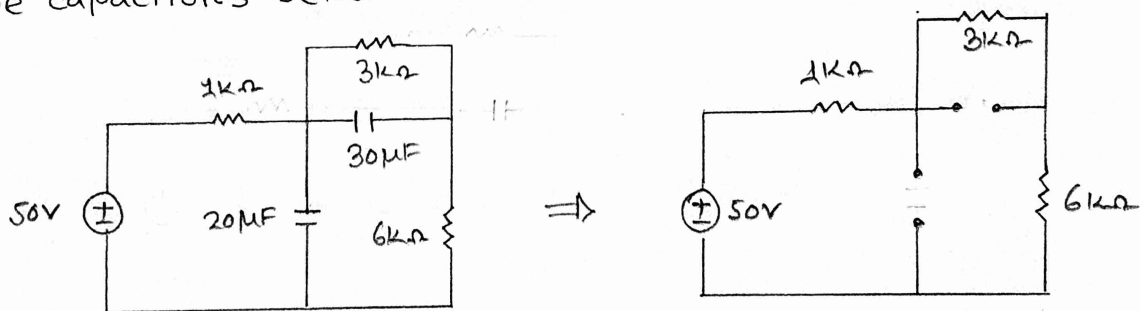
$$\therefore V_{4k\Omega} = 2 \times 4 = 8V$$

$$\begin{aligned} \therefore E_{2mF} &= \frac{1}{2} \times C_1 \times V_{2k\Omega}^2 \\ &= \frac{1}{2} \times 2 \times 10^{-3} \times 4^2 = 0.016 J \end{aligned}$$

$$\begin{aligned} \therefore E_{4mF} &= \frac{1}{2} \times C_2 \times V_{4k\Omega}^2 \\ &= \frac{1}{2} \times 4 \times 10^{-3} \times 8^2 = 0.128 J \end{aligned}$$

(Ans.)

Problem 09: Under dc conditions find the energy stored in the capacitors below.



Solution :

$$R_{eq} = 6 + 3 + 1 = 10 k\Omega$$

$$\therefore I = V / R_{eq} = 50 / 10 = 5 \text{ mA}$$

$$\therefore V_{3k\Omega} = (3 \times 5) = 15V$$

$$\therefore V_{(3+6)k\Omega} = (9 \times 5) = 45V$$

$$\therefore W_{30\mu F} = \frac{1}{2} \times 30 \times 10^{-6} \times V_{3k\Omega}^2 = \frac{1}{2} \times 30 \times 10^{-6} \times 15^2 = 3.375 \times 10^{-3} J$$

$$\therefore W_{20\mu F} = \frac{1}{2} \times 20 \times 10^{-6} \times V_{(3+6)k\Omega}^2 = \frac{1}{2} \times 20 \times 10^{-6} \times 45^2 = 20.25 \times 10^{-3} J$$

(Ans.)

Here,

$$C_1 = 2 \times 10^{-3} F$$

$$C_2 = 4 \times 10^{-3} F$$

□ All capacitors replaced by open circuit under dc condition.

Problem 10: Calculate the energy stored in the capacitor of the circuit in fig below under dc condition.

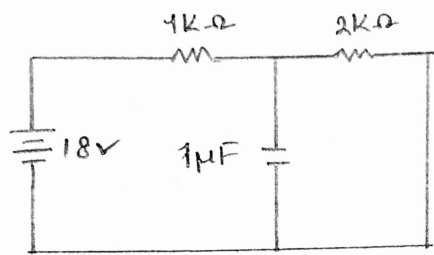


Fig: a

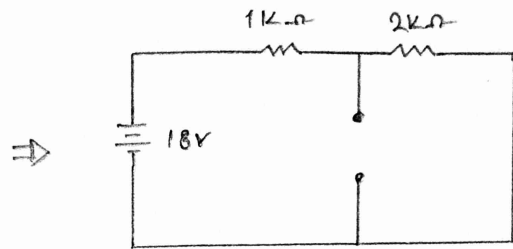


Fig: b

Solⁿ: Under dc condition we replace the capacitor with open circuit.

Hence, the voltage drop in $2k\Omega$ resistor is the voltage drop of $1\mu F$ capacitor.

Given,
 $E = 18V$

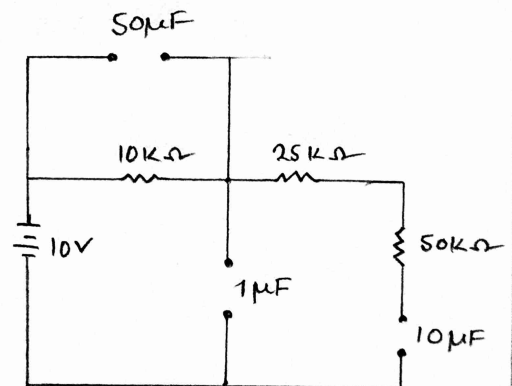
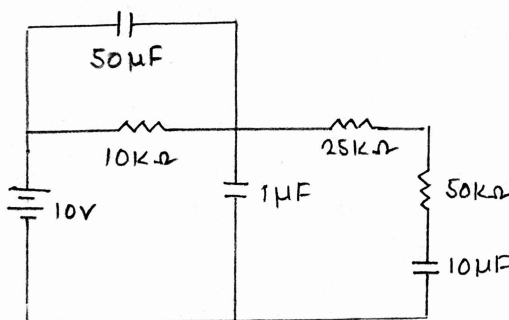
$$\therefore R_{eq} = 2 + 1 = 3k\Omega$$

$$\therefore I = \frac{E}{R_{eq}} = \frac{18}{3} = 6mA$$

$$\therefore V_{2k\Omega} = 6 \times 2 = 12V$$

$$\therefore W_{1\mu F} = \frac{1}{2} \times 1 \times 10^{-6} \times 12^2 = 7.2 \times 10^{-5} J \quad (\text{Ans.})$$

Problem 11: Calculate the energy stored in the below circuit under dc condition.



Solⁿ: Under dc condition we replace the capacitors with open circuit.

Since the circuit becomes open circuit there will be no current flow. so, by the terminal we can determine,

$$V_{50\mu F} = V_{10k\Omega} = 0V$$

$$V_{1\mu F} = 10V$$

$$V_{10\mu F} = 10V$$

$$\therefore W_{50\mu F} = \frac{1}{2} \times 50 \times 10^{-6} \times 0 = 0J$$

$$\therefore W_{1\mu F} = \frac{1}{2} \times 1 \times 10^{-6} \times 10^2 = 50 \times 10^{-6}J$$

$$\therefore W_{10\mu F} = \frac{1}{2} \times 10 \times 10^{-6} \times 10^2 = 500 \times 10^{-6}J$$

(Ans.)

problem 13: Find the voltage across and the charge on capacitor C and after it has charged upto its final value.

Solⁿ.

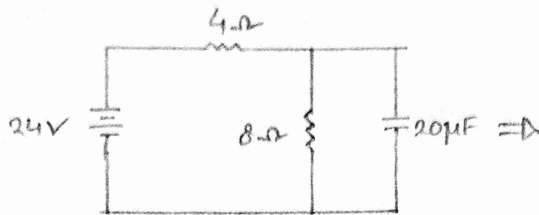


Fig: a

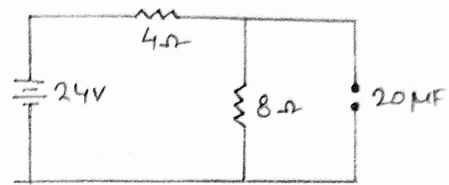


Fig: b

Under dc condition we replace the capacitor with open circuit.

So, voltage drop of $20\mu\text{F}$ is same as voltage drop in 8Ω .

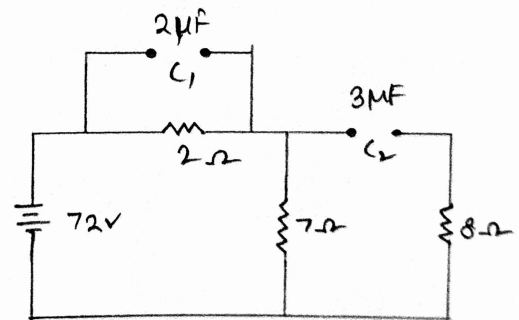
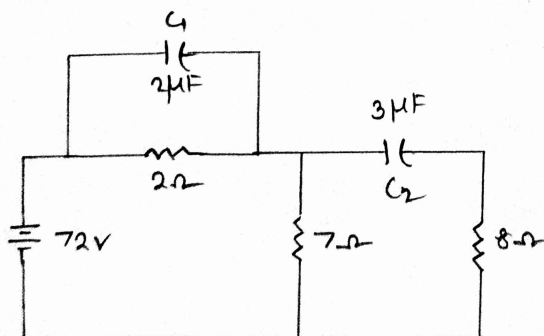
$$\therefore V_{8\Omega} = \frac{8}{8+4} \times 24 = 16 \text{ V}$$

We know,

$$Q = CV = 20 \times 10^{-6} \times 16 = 320 \times 10^{-6} \text{ C}$$

(Ans.)

Problem 14: Find the voltage across and the charge on each capacitor of the network after each has charged up to its final value.



under dc condition, we replace the capacitor with open circuit.

So,

$$\therefore V_{2\mu F} = V_{2\Omega} = \frac{2}{7+2} \times 72 = 16 \text{ V}$$

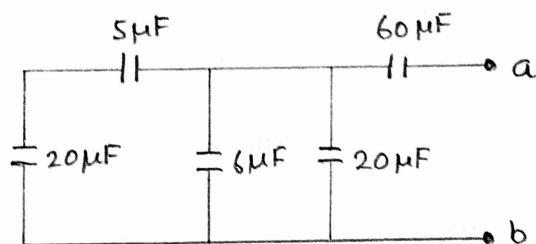
$$\therefore V_{3\mu F} = V_{7\Omega} = \frac{7}{7+2} \times 72 = 56 \text{ V}$$

$$\therefore Q_{2\mu F} = C_1 \times V_{2\mu F} = 2 \times 10^{-6} \times 16 = 32 \times 10^{-6} \text{ C}$$

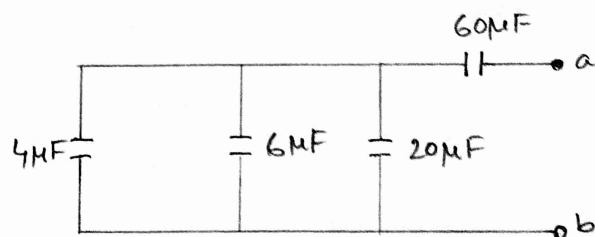
$$\therefore Q_{3\mu F} = C_2 \times V_{3\mu F} = 3 \times 10^{-6} \times 56 = 168 \times 10^{-6} \text{ C}$$

(Ans.)

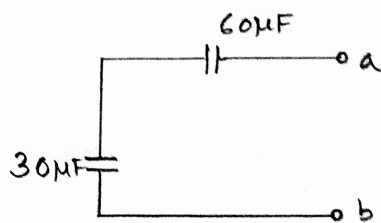
Problem 15: Find the equivalent capacitance seen between terminals a and b.



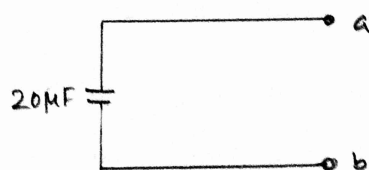
$$5\mu F + 20\mu F \\ = \left(\frac{1}{5} + \frac{1}{20}\right) = 4\mu F$$



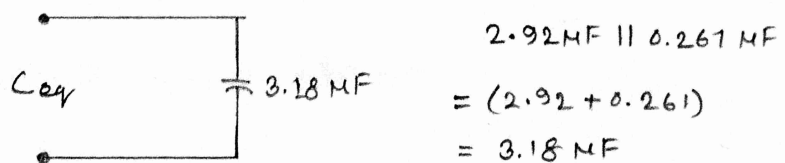
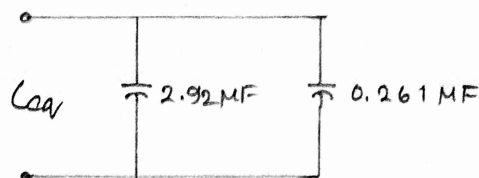
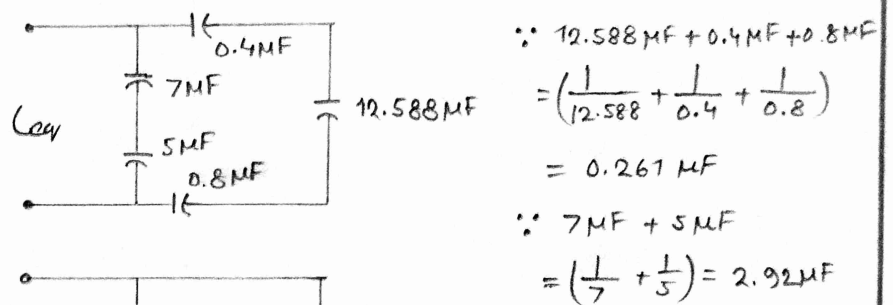
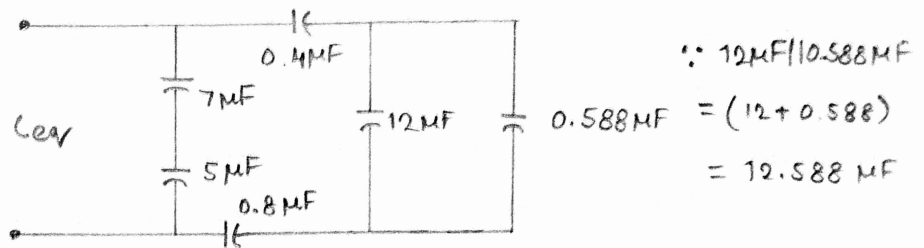
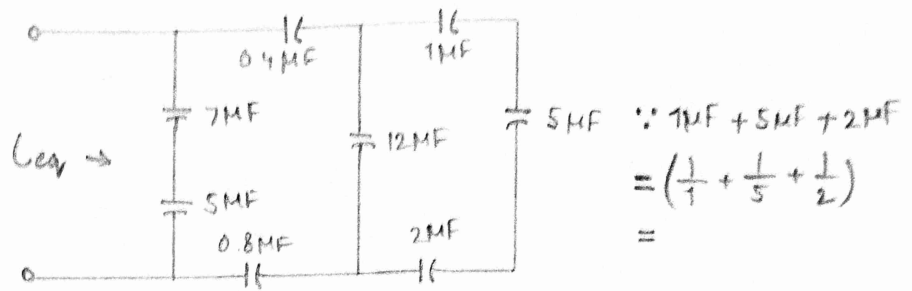
$$4\mu F \parallel 6\mu F \parallel 20\mu F \\ = (4 + 6 + 20) \\ = 30\mu F$$



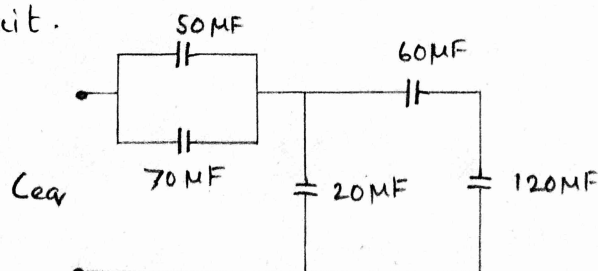
$$60\mu F + 30\mu F \\ = \left(\frac{1}{60} + \frac{1}{30}\right) \\ = 20\mu F$$



Problem 16: Find C_{eq} for the following network.



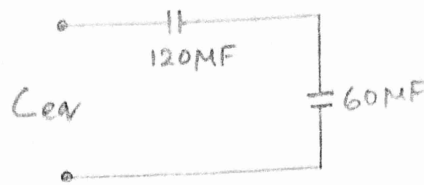
problem 17: Find the equivalent capacitance seen at the terminal of the circuit.



$$60\mu F + 120\mu F$$

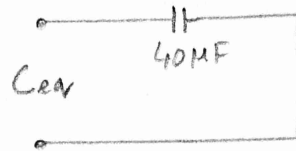
$$= \left(\frac{1}{60} + \frac{1}{120} \right)$$

$$= 40\mu F$$



$$\therefore 50\text{MF} \parallel 70\text{MF} \\ = (50 + 70) = 120\text{MF}$$

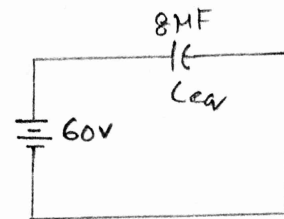
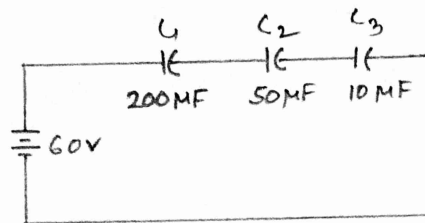
$$\therefore 40\text{MF} \parallel 20\text{MF} \\ = (40 + 20) = 60\text{MF}$$



$$\therefore 120\text{MF} + 60\text{MF} \\ = \left(\frac{1}{120} + \frac{1}{60}\right) = 40\text{MF}$$

Problem 18: Find the total capacitance, Determine the charge on each plate, find the voltage across each capacitor. for the following figure.

Soln.



$$a) \therefore C_{eq} = \left(\frac{1}{200} + \frac{1}{50} + \frac{1}{10}\right) = 8\text{MF} \quad (\text{Ans})$$

$$b) \text{ Given, } V = 60\text{V}$$

$$\therefore q_T = q_1 = q_2 = q_3$$

$$\therefore q_T = C_{eq} V = 8 \times 60 = 480\text{MF} \quad (\text{Ans})$$

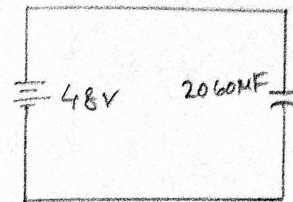
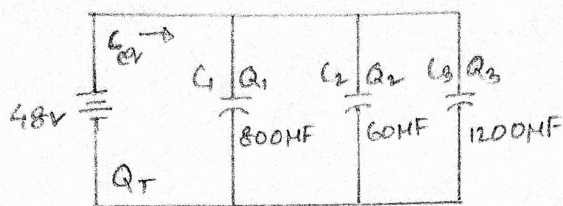
$$c) \therefore V_1 = \frac{q_1}{C_1} = \frac{480 \times 10^{-6}}{200 \times 10^{-6}} = 2.4\text{V}$$

$$\therefore V_2 = \frac{q_2}{C_2} = \frac{480 \times 10^{-6}}{50 \times 10^{-6}} = 9.6\text{V}$$

$$\therefore V_3 = \frac{q_3}{C_3} = \frac{480 \times 10^{-6}}{10 \times 10^{-6}} = 48\text{V} \quad (\text{Ans})$$

Problem 19: Find the total capacitance, determine the charge on each plate, find the total charge.

Solⁿ:



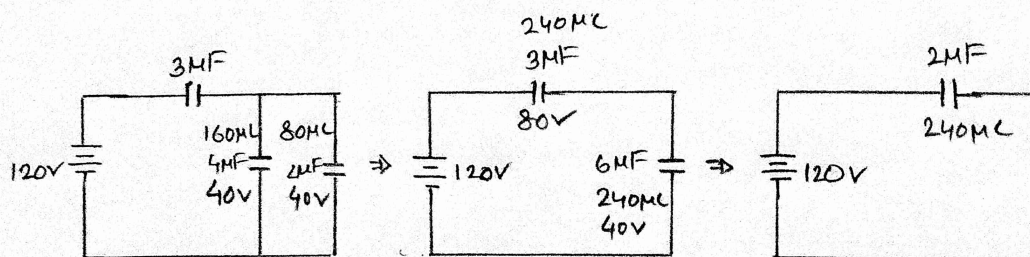
a) $C_{eq} = C_1 + C_2 + C_3 = 800\text{MF} + 60\text{MF} + 1200\text{MF} = 2060\text{MF}$
(Ans)

b) $Q_1 = C_1 V = 800 \times 10^{-6} \times 48 = 38.4 \times 10^{-3} \text{C}$
 $Q_2 = C_2 V = 60 \times 10^{-6} \times 48 = 2.88 \times 10^{-3} \text{C}$
 $Q_3 = C_3 V = 1200 \times 10^{-6} \times 48 = 57.6 \times 10^{-3} \text{C}$
(Ans)

c) $Q_T = Q_1 + Q_2 + Q_3 = 38.4 \times 10^{-3} + 2.88 \times 10^{-3} + 57.6 \times 10^{-3}$
 $= 98.88 \times 10^{-3} \text{C}$
(Ans)

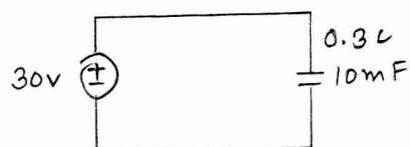
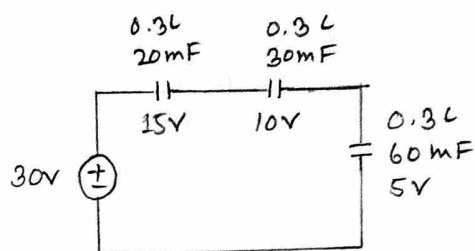
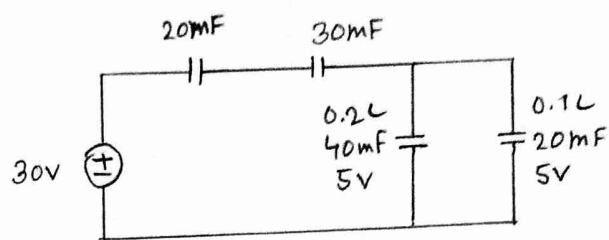
Problem 20: Find the voltage across and the charge on each capacitor for the network below.

Solⁿ:



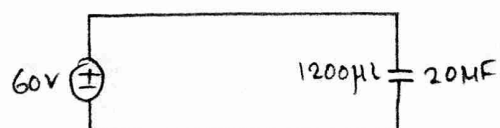
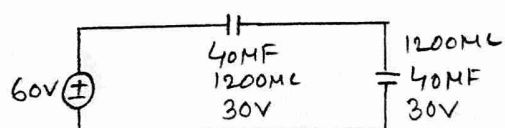
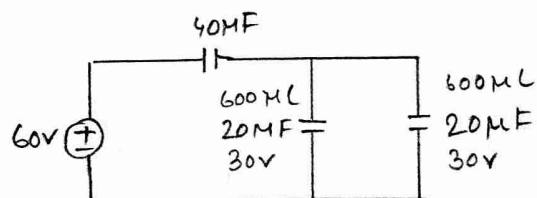
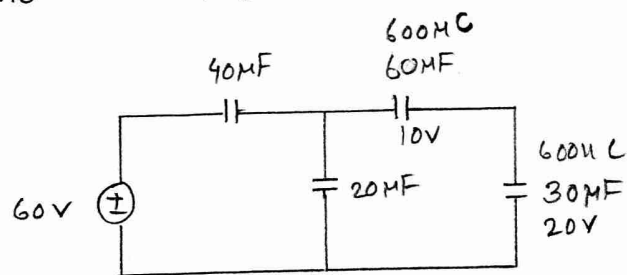
Problem 21: Find the voltage across each capacitor.

Solⁿ.

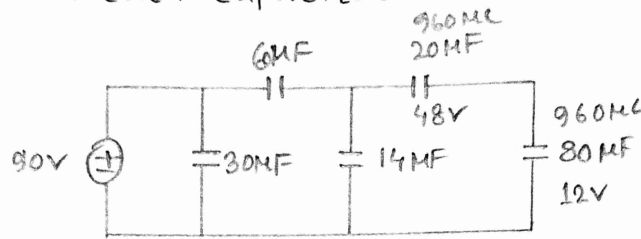


Problem 22: Find the voltage across each of the capacitors.

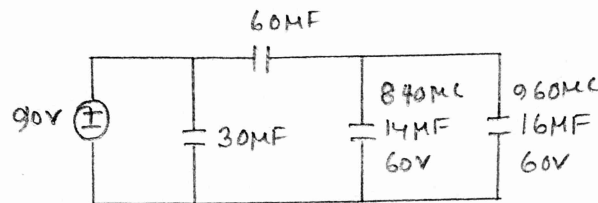
Solⁿ.



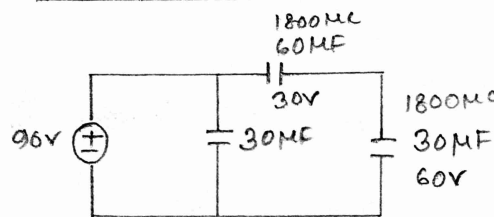
Problem 23: Find the voltage across each capacitor and the energy stored in each capacitor.



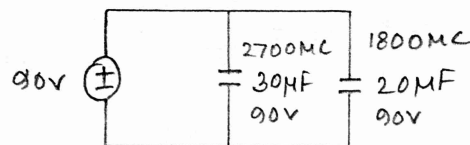
$$\begin{aligned} \therefore 20\mu F + 80\mu F \\ &= \left(\frac{1}{20} + \frac{1}{80} \right) \\ &= 16\mu F \end{aligned}$$



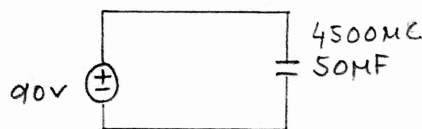
$$\begin{aligned} \therefore 16\mu F \parallel 14\mu F \\ &= (16 + 14) = 30\mu F \end{aligned}$$



$$\begin{aligned} \therefore 60\mu F + 30\mu F \\ &= \left(\frac{1}{60} + \frac{1}{30} \right) = 20\mu F \end{aligned}$$



$$\begin{aligned} \therefore 30\mu F \parallel 20\mu F \\ &= (30 + 20) = 50\mu F \end{aligned}$$



$$\therefore V_{30\mu F} = 90V \quad \therefore V_{60\mu F} = 30V \quad \therefore V_{14\mu F} = 60V \quad \therefore V_{20\mu F} = 48V$$

$$\therefore V_{80\mu F} = 12V$$

$$\therefore W_{30\mu F} = \frac{1}{2} \times 30 \times 10^{-6} \times 90^2 = 121.5 \times 10^{-3} \text{ J}$$

$$\therefore W_{60\mu F} = \frac{1}{2} \times 60 \times 10^{-6} \times 30^2 = 27 \times 10^{-3} \text{ J}$$

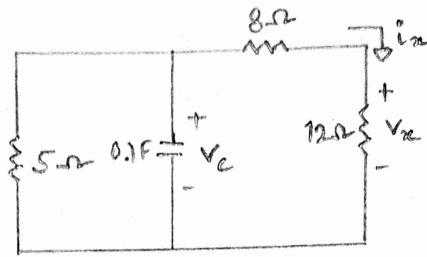
$$\therefore W_{14\mu F} = \frac{1}{2} \times 14 \times 10^{-6} \times 60^2 = 25.2 \times 10^{-3} \text{ J}$$

$$\therefore W_{20\mu F} = \frac{1}{2} \times 20 \times 10^{-6} \times 48^2 = 23.04 \times 10^{-3} \text{ J}$$

$$\therefore W_{80\mu F} = \frac{1}{2} \times 80 \times 10^{-6} \times 12^2 = 5.76 \times 10^{-3} \text{ J}$$

(Ans.)

Problem 24: Let $V_C(0) = 15\text{ V}$, Find V_C , V_X and i_X for $t \geq 0$



Solⁿ:

$$\therefore R_{eq} = \left(\frac{1}{5} + \frac{1}{8+12} \right) = 4\Omega \quad \left| \begin{array}{l} C = 0.1\text{ F (Given)} \\ V_0 = 15\text{ V} \end{array} \right.$$

$$\therefore \tau = R_{eq} C = 4 \times 0.1 = 0.4\text{ s}$$

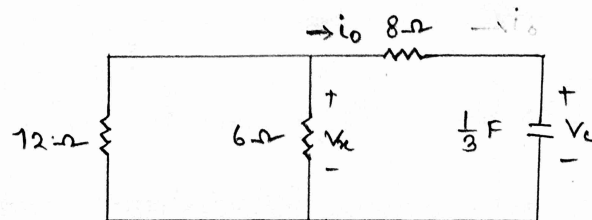
$$\therefore V_C = V_0 e^{-t/\tau} = 15 e^{-t/0.4} = 15 e^{-2.5t}\text{ V}$$

$$\therefore V_X = \frac{12}{12+8} \cdot 15 e^{-2.5t} = 9 e^{-2.5t}\text{ V}$$

$$\therefore i_X = V_X/12 = \frac{9 e^{-2.5t}}{12} = 0.75 e^{-2.5t}\text{ A}$$

(Ans)

Problem 25: Let $V_C(0) = 45\text{ V}$. Determine V_C , V_X and i_0 for $t \geq 0$



Solⁿ:

$$R_{eq} = \left(\frac{1}{12} + \frac{1}{6} \right)^{-1} + 8 = 4 + 8 = 12\Omega$$

Given, $C = \frac{1}{3}\text{ F}$

$$V_C(0) = V_0 = 45\text{ V}$$

$$\tau = R_{eq} C = 12 \times \frac{1}{3} = 4\text{ s}$$

$$\therefore V_C = V_0 e^{-t/\tau} = 45 e^{-t/4}$$

$$\therefore R_p = \left(\frac{1}{12} + \frac{1}{6} \right)^{-1} = 4\Omega$$

$$\therefore V_x = \frac{R_p}{R_{eq}} \times V_c = \frac{4}{12} \times 45e^{-t/4} = 15e^{-t/4}$$

$$\therefore i_o = \frac{45e^{-t/4}}{R_{eq}} = \frac{45e^{-t/4}}{12} = 3.75e^{-t/4} \text{ A } (\leftarrow)$$

$$= -3.75e^{-t/4} \text{ A } (\rightarrow)$$

(Ans.)

Problem 26: The switch in the circuit has been closed for a long time, and it is opened at $t=0$. Find $v(t)$ for $t \geq 0$. Calculate the initial energy stored in the capacitor.

Soln:

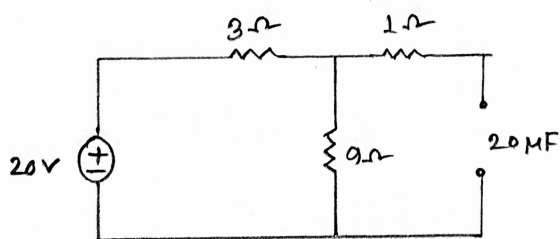
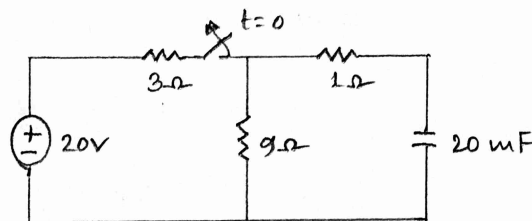


Fig: 1

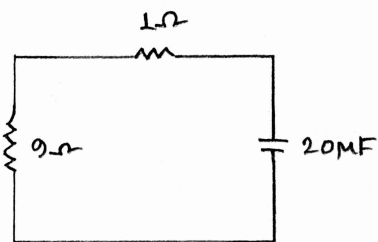


Fig: 2

After fully charged in dc condition capacitor acts as an open circuit. So the voltage drop across 9Ω is equal to voltage drop in $20\mu\text{F}$ capacitor. (Fig: 1)

$$\therefore V_0 = \frac{9}{3+9} \times 20 = 15\text{V}$$

In discharged phase, circuit will follow similarly 'Fig: 02'.

$$\therefore R_{eq} = (9+1) = 10\Omega$$

We know,

$$\tau = R_{eq} C$$

$$= 10 \times 20 \times 10^{-3}$$

$$= 200 \times 10^{-3} \text{ s}$$

$$\begin{aligned}\therefore V_c &= V_0 e^{-\frac{t}{\tau}} \\ &= 15 \cdot e^{-\frac{t}{200 \times 10^{-3}}} \\ &= 15 e^{-5t}\end{aligned}$$

$$\therefore W_c(t) = \frac{1}{2} C V_c^2(t)$$

$$= \frac{1}{2} \times 20 \times 10^{-3} \times (15 e^{-5t})^2$$

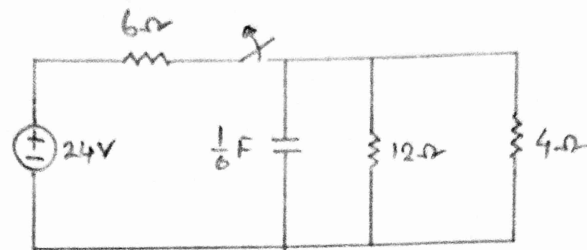
$$= \frac{1}{2} \times 20 \times 10^{-3} \times (15 e^{-5 \times 0})^2 \quad [\because t=0]$$

$$= \frac{1}{2} \times 20 \times 10^{-3} \times (15)^2$$

$$= 2.25 \text{ J}$$

(Ans.)

Problem 27: If the switch opens at $t=0$, find $v(t)$ for $t \geq 0$ and $W_C(0)$.



Solⁿ:

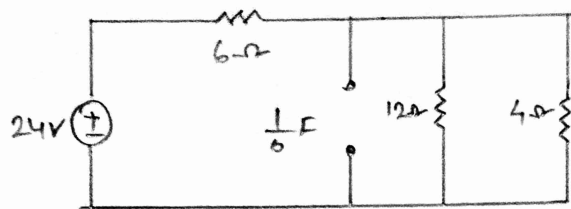


Fig: 1

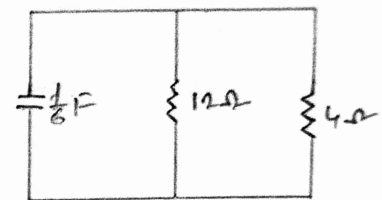


Fig: 2

After fully charged in dc condition a capacitor acts as an open circuit.

Hence, $\frac{1}{6}$ F capacitor's voltage drop is same equivalent resistor of 12Ω and 4Ω .

$$\therefore R_p = \left(\frac{1}{12} + \frac{1}{4} \right)^{-1} = 3\Omega$$

$$\therefore V_0 = \frac{3}{3+6} \times 24 = 8V$$

$$\therefore \tau = R_p C = 3 \times \frac{1}{6} = 0.5 \text{ s}$$

$$\therefore V_C = V_0 e^{-t/\tau} = 8 e^{-t/0.5} = 8 e^{-2t} \text{ V}$$

$$\therefore W_C(0) = \frac{1}{2} C V_C^2(0)$$

$$= \frac{1}{2} \times \frac{1}{6} \times (8 e^{-2 \times 0})^2$$

[$\because t=0$]

$$= \frac{1}{2} \times \frac{1}{6} \times 8^2 = 5.33 \text{ J}$$

(Ans.)

Problem 28: Find i from the following circuit.

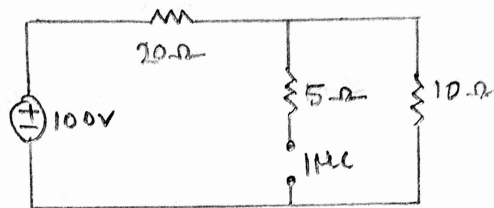
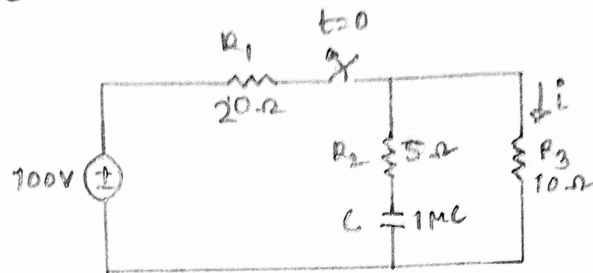


Fig: 01

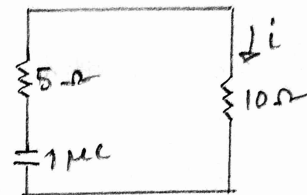


Fig: 2

In fully charged phase a capacitor acts as an open circuit in dc configuration. (Fig: 01)

$$\therefore V_0 = \frac{10}{20+10} \times 100 = 33.34 \text{ V}$$

From fig (2) we can obtain equivalent resistance as the discharging phase of the capacitor.

$$\therefore R_{eq} = 5 + 10 = 15 \Omega$$

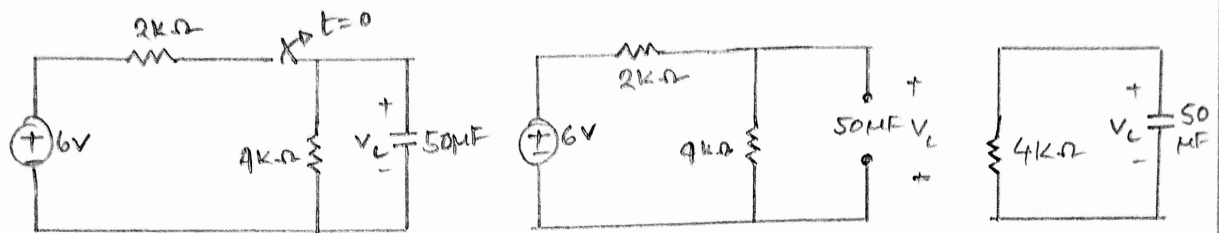
$$\therefore \tau = R_{eq} C = 15 \times 1 \times 10^{-6} = 15 \times 10^{-6} \text{ s}$$

$$\therefore V_C = V_0 e^{-t/\tau} = 33.34 e^{-t/15 \times 10^{-6}} = 33.34 e^{-66.67 \times 10^3 t}$$

$$\therefore i = \frac{V_C}{10} = \frac{33.34 e^{-66.67 \times 10^3 t}}{10} = 3.334 e^{-66.67 \times 10^3 t} \text{ A}$$

(Ans.)

Problem 29: The switch opens at $t=0$. Find V_C .



Soln:

under dc condition capacitor acts as open circuit.

$$\therefore V_0 = V_{4k\Omega} = \frac{4}{2+4} \times 6 = 4V$$

$$\therefore R_{eq} = 4k\Omega$$

$$\therefore \tau = 4 \times 10^3 \times 50 \times 10^{-6} s$$

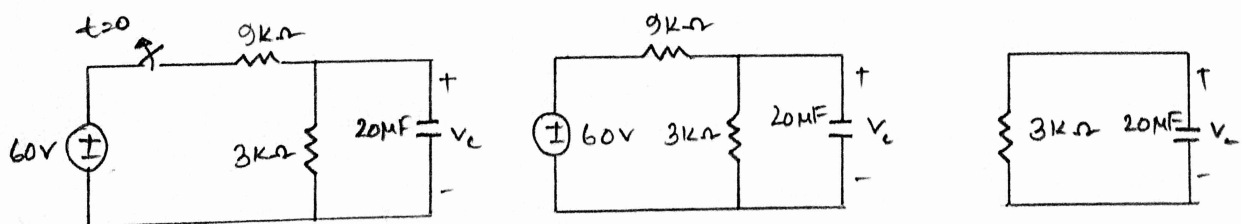
$$= 0.2 s$$

$$[\because \tau = R_{eq} C]$$

$$\therefore V_C = V_0 e^{-\frac{t}{\tau}} = 4 e^{-\frac{t}{0.2}} = 4 e^{-5t} V$$

(Ans.)

Problem 30: Find $V_C(t)$ for $t > 0$. Determine the time necessary for the capacitor voltage to decay to one-third of its value at $t=0$.



$$V_0 = V_{3k\Omega} = \frac{3}{3+9} \times 60 = 15V$$

$$\therefore R_{eq} = 3 \times 10^3 \Omega$$

$$\therefore \tau = R_{eq} C = 3 \times 10^3 \times 20 \times 10^{-6} s$$

$$= 0.06 s$$

$$\therefore V_C = V_0 e^{-t/\tau} = 15 e^{-t/0.06} = 15 e^{-16.67 t}$$

From the question,

$$\frac{15}{3} = 15 e^{-16.67 t}$$

$$\Rightarrow 5 = 15 e^{-16.67 t}$$

$$\Rightarrow e^{-16.67 t} = \frac{1}{3}$$

$$\Rightarrow \ln(e^{-16.67 t}) = \ln\left(\frac{1}{3}\right)$$

$$\Rightarrow -16.67 t = \ln\left(\frac{1}{3}\right)$$

$$\therefore t = \frac{\ln(1/3)}{-16.67}$$

$$= 65.92 \times 10^{-3} \text{ s}$$

(Ans.)

Conclusion:

The comprehensive analysis of 29 capacitor problems has successfully demonstrated the application of fundamental principles in both DC and AC circuits. Through methodical problem solving, we explored various aspects of capacitor behavior, including equivalent capacitance, energy storage, voltage drop, current and time dependent characteristics. This systematic approach has reinforced core concepts while developing essential analytical skills for electrical engineering applications.